

Conditional independence: the algebra behind the models

“Algebraic & Geometric Methods in Statistics”

Sonja Petrović
Created for Math/Stat 561

Jan 22, 2026.

Objective

Understand how to translate conditional independence statements into polynomials, and what these polynomials mean.

Recall conditional independence (CI) from Lecture 3

Definition 4.1.2. Let $A, B, C \subseteq [m]$ be pairwise disjoint. The random vector X_A is *conditionally independent of X_B given X_C* if and only if

$$f_{A \cup B | C}(x_A, x_B | x_C) = f_{A | C}(x_A | x_C) \cdot f_{B | C}(x_B | x_C)$$

for all x_A, x_B , and x_C . The notation $X_A \perp\!\!\!\perp X_B | X_C$ is used to denote that the random vector X satisfies the conditional independence statement that X_A is conditionally independent of X_B given X_C . This is often further abbreviated to $A \perp\!\!\!\perp B | C$.

Figure 1: Source: Algebraic statistics, Seth Sullivant, AMS-GSM book

Real life examples¹

Reminder: The *conditional probability* of A given B is represented by $P(A|B)$. The random variables A and B are said to be **independent** if $P(A) = P(A|B)$ (or alternatively if $P(A,B) = P(A) P(B)$).

Example 1

Suppose Norman and Martin each toss **separate coins**.

- Let A represent the random variable “Norman’s toss outcome”, and B represent the random variable “Martin’s toss outcome”.
- Both A and B have two possible values (Heads and Tails).
- It would be uncontroversial to assume that A and B are independent.

Evidence about B will not change our belief in A.

¹Credit: Normal Fenton

Example 2

Now suppose both Martin and Norman toss **the same coin**.

- Again $A = \text{"Norman's toss outcome"}$, and $B = \text{"Martin's toss outcome"}$.
- Assume also that **there is a possibility that the coin is biased towards heads** but we do **not** know this for certain.
- In this case A and B are **not** independent.

Example: observing $B = \text{Heads}$ causes us to increase our belief in $A = \text{Heads}$! So $P(a|b) > P(b)$ in the case when $a = \text{Heads}$ and $b = \text{Heads}$.

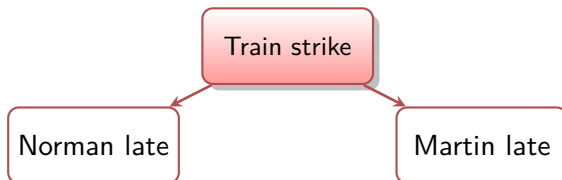
- RVs A and B are **both dependent on a separate random variable C** , “the coin is biased towards Heads” (which has the values True or False).
- Although A and B are not independent, it turns out that once we know for certain the value of C then any evidence about B cannot change our belief about A .

Specifically: $P(A|C) = P(A|B, C)$, so $CI\ A \perp\!\!\!\perp B|C$ holds.

Example 3: $A \perp\!\!\!\perp B | C$ see full gif

In many real life situations variables which are believed to be *independent* are actually *only independent conditional* on some other variable.

- Norman and Martin live on opposite sides of the City
- Norman takes the train to work. Martin drives.
- Random variables: A = “Norman late” , B = “Martin late” (true/false)
- $A \perp\!\!\!\perp B$??
 - are you sure? what about fuel shortage?
 - what about ... more traffic on the road due to a train strike?
- Let C = “train strike”.
- Clearly $P(A)$ will increase if C is true; but $P(B)$ will also increase because of extra traffic on the roads.



Example $\mapsto$ homework 2 [an actual HW problem!]

Discussion of the setup. ****[Whiteboard illustration.]****

Consider three **binary** random variables X_1, X_2, X_3 , with joint probabilities $P(X_1 = i, X_2 = j, X_3 = 0) = P_{i,j}^{(X_3=0)}$ and $P(X_1 = i, X_2 = j, X_3 = 1) = P_{i,j}^{(X_3=1)}$, with:

$$P^{(X_3=0)} := \begin{pmatrix} 0.05 & 0.15 \\ 0.075 & 0.225 \end{pmatrix}, P^{(X_3=1)} := \begin{pmatrix} 0.125 & 0.125 \\ 0.125 & 0.125 \end{pmatrix}.$$

$\rightarrow$ This is a $2 \times 2 \times 2$ table, similar to the Berkeley admissions example in lecture3 handout (!). $\leftarrow$

- Find the marginal distribution P_{X_1} of X_1 . (Recall that in the discrete case, integration is substituted by summation.)
- Find the conditional distribution $P_{X_2, X_3 | X_1}$ of (X_2, X_3) given X_1 .
- Is X_2 conditionally independent of X_3 given X_1 ?
- Is X_1 conditionally independent of X_2 given X_3 ?

The CI statement is a polynomial in the model probabilities!

Proposition (4.1.6.) & Definition (4.1.7.)

If X is a discrete random vector $X = (X_1, \dots, X_m)$, then the CI statement $X_A \perp\!\!\!\perp X_B | X_C$ is equivalent to

$$p_{i_A, i_B, i_C, +} \cdot p_{j_A, j_B, i_C, +} - p_{i_A, j_B, i_C, +} \cdot p_{j_A, i_B, i_C, +} = 0$$

for all possible states of the variables i_A, j_A, i_B, j_B , and i_C .

The **CI ideal** $I_{A \perp\!\!\!\perp B | C}$ is the set of polynomials **generated** by **all** quadratic polynomials above.

The CI statement is a polynomial in the model probabilities!

Proposition (4.1.6.) & Definition (4.1.7.)

If X is a discrete random vector $X = (X_1, \dots, X_m)$, then the CI statement $X_A \perp\!\!\!\perp X_B | X_C$ is equivalent to

$$p_{i_A, i_B, i_C, +} \cdot p_{j_A, j_B, i_C, +} - p_{i_A, j_B, i_C, +} \cdot p_{j_A, i_B, i_C, +} = 0$$

for all possible states of the variables i_A, j_A, i_B, j_B , and i_C .

The **CI ideal** $I_{A \perp\!\!\!\perp B | C}$ is the set of polynomials **generated** by **all** quadratic polynomials above.

- Week 1: we wrote the 3-step binary Markov chain model as a (semi)algebraic set: the set of probability distributions satisfying polynomial equations (and inequalities).
- ... **all** polynomials? how many are there?
- ... **“generated”**?



Advanced HW (on hw2)

Prove proposition 4.1.6. The outline of the proof is in the book.

Example

Let X_1, X_2, X_3, X_4 be four discrete random variables with the following state spaces: $X_1 \subset \{1, 2, 3\}$, $X_2, X_3, X_4 \subset \{1, 2\}$.

- Interpret: X_1 = gender (M/F/other), X_2 = short hair (1=y/2=n), X_3 = likes soccer (y/n), X_4 = from Brazil (y/n).

$$X_1 \perp\!\!\!\perp X_2 | X_3 \iff$$

Advanced HW (on hw2)

Prove proposition 4.1.6. The outline of the proof is in the book.

Example

Let X_1, X_2, X_3, X_4 be four discrete random variables with the following state spaces: $X_1 \subset \{1, 2, 3\}$, $X_2, X_3, X_4 \subset \{1, 2\}$.

- Interpret: X_1 = gender (M/F/other), X_2 = short hair (1=y/2=n), X_3 = likes soccer (y/n), X_4 = from Brazil (y/n).

$$X_1 \perp\!\!\!\perp X_2 | X_3 \iff p_{1,1,1,+} \cdot p_{2,2,1,+} - p_{1,2,1,+} \cdot p_{2,1,1,+} = 0$$

Advanced HW (on hw2)

Prove proposition 4.1.6. The outline of the proof is in the book.

Example

Let X_1, X_2, X_3, X_4 be four discrete random variables with the following state spaces: $X_1 \subset \{1, 2, 3\}$, $X_2, X_3, X_4 \subset \{1, 2\}$.

- Interpret: X_1 = gender (M/F/other), X_2 = short hair (1=y/2=n), X_3 = likes soccer (y/n), X_4 = from Brazil (y/n).

$$X_1 \perp\!\!\!\perp X_2 | X_3 \iff p_{1,1,1,+} \cdot p_{2,2,1,+} - p_{1,2,1,+} \cdot p_{2,1,1,+} = 0$$

$$p_{1,1,1,+} \cdot p_{2,3,1,+} - p_{1,3,1,+} \cdot p_{2,1,1,+} = 0$$

$$p_{1,2,1,+} \cdot p_{2,3,1,+} - p_{1,3,1,+} \cdot p_{2,2,1,+} = 0$$

...

Advanced HW (on hw2)

Prove proposition 4.1.6. The outline of the proof is in the book.

Example

Let X_1, X_2, X_3, X_4 be four discrete random variables with the following state spaces: $X_1 \subset \{1, 2, 3\}$, $X_2, X_3, X_4 \subset \{1, 2\}$.

- Interpret: X_1 = gender (M/F/other), X_2 = short hair (1=y/2=n), X_3 = likes soccer (y/n), X_4 = from Brazil (y/n).

$$X_1 \perp\!\!\!\perp X_2 | X_3 \iff p_{1,1,1,+} \cdot p_{2,2,1,+} - p_{1,2,1,+} \cdot p_{2,1,1,+} = 0$$

$$p_{1,1,1,+} \cdot p_{2,3,1,+} - p_{1,3,1,+} \cdot p_{2,1,1,+} = 0$$

$$p_{1,2,1,+} \cdot p_{2,3,1,+} - p_{1,3,1,+} \cdot p_{2,2,1,+} = 0$$

...

$$p_{1,1,2,+} \cdot p_{2,2,2,+} - p_{1,2,2,+} \cdot p_{2,1,2,+} = 0 \text{ and more! !!}$$

- **And** all of these +s mean, e.g., $p_{1,1,1,+} = p_{1,1,1,1} + p_{1,1,1,2}$.

Question

Is there an efficient way of (1) encoding these polynomials and (2) generating them for a simple example??

Example Macaulay2 code

```
Macaulay2, version 1.18
i1 : loadPackage "GraphicalModels";
i2 : R = markovRing (3,2,2,2);
o2 = R
o2 : PolynomialRing
i3 : rvNames = {gender, hair, soccer, brazil}
o3 = {gender, hair, soccer, brazil}
o3 : List
i4 : CIstatements = { {{gender},{hair},{soccer}} }
-- this says gender indep. of hair given soccer
o4 = {{{gender},{hair},{soccer}}}}
o4 : List
i5 : conditionalIndependenceIdeal(R, CIstatements, rvNames)
```

You can compute this online yourself.. The lines starting with “i” are input lines that you type into the editor to execute them.

$$\begin{aligned}
& \text{ideal}(-p_{1,2,1,1}p_{2,1,1,1} - p_{1,2,1,2}p_{2,1,1,1} - p_{1,2,1,1}p_{2,1,1,2} - p_{1,2,1,2}p_{2,1,1,2} + \\
& p_{1,1,1,1}p_{2,2,1,1} + p_{1,1,1,2}p_{2,2,1,1} + p_{1,1,1,1}p_{2,2,1,2} + p_{1,1,1,2}p_{2,2,1,2}, \\
& -p_{1,2,1,1}p_{3,1,1,1} - p_{1,2,1,2}p_{3,1,1,1} - p_{1,2,1,1}p_{3,1,1,2} - p_{1,2,1,2}p_{3,1,1,2} + \\
& p_{1,1,1,1}p_{3,2,1,1} + p_{1,1,1,2}p_{3,2,1,1} + p_{1,1,1,1}p_{3,2,1,2} + p_{1,1,1,2}p_{3,2,1,2}, \\
& -p_{2,2,1,1}p_{3,1,1,1} - p_{2,2,1,2}p_{3,1,1,1} - p_{2,2,1,1}p_{3,1,1,2} - p_{2,2,1,2}p_{3,1,1,2} + \\
& p_{2,1,1,1}p_{3,2,1,1} + p_{2,1,1,2}p_{3,2,1,1} + p_{2,1,1,1}p_{3,2,1,2} + p_{2,1,1,2}p_{3,2,1,2}, \\
& -p_{1,2,2,1}p_{2,1,2,1} - p_{1,2,2,2}p_{2,1,2,1} - p_{1,2,2,1}p_{2,1,2,2} - p_{1,2,2,2}p_{2,1,2,2} + \\
& p_{1,1,2,1}p_{2,2,2,1} + p_{1,1,2,2}p_{2,2,2,1} + p_{1,1,2,1}p_{2,2,2,2} + p_{1,1,2,2}p_{2,2,2,2}, \\
& -p_{1,2,2,1}p_{3,1,2,1} - p_{1,2,2,2}p_{3,1,2,1} - p_{1,2,2,1}p_{3,1,2,2} - p_{1,2,2,2}p_{3,1,2,2} + \\
& p_{1,1,2,1}p_{3,2,2,1} + p_{1,1,2,2}p_{3,2,2,1} + p_{1,1,2,1}p_{3,2,2,2} + p_{1,1,2,2}p_{3,2,2,2}, \\
& -p_{2,2,2,1}p_{3,1,2,1} - p_{2,2,2,2}p_{3,1,2,1} - p_{2,2,2,1}p_{3,1,2,2} - p_{2,2,2,2}p_{3,1,2,2} + \\
& p_{2,1,2,1}p_{3,2,2,1} + p_{2,1,2,2}p_{3,2,2,1} + p_{2,1,2,1}p_{3,2,2,2} + p_{2,1,2,2}p_{3,2,2,2})
\end{aligned}$$

Class work:

Determine why these are correct. [worksheet 2](#).

Homework 1, problem 3 [due today]

Example 3.1.6 (Marginal independence). The (marginal) independence statement $X_1 \perp\!\!\!\perp X_2$, or equivalently, $X_1 \perp\!\!\!\perp X_2 \mid X_\emptyset$, amounts to saying that the matrix

$$\begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1r_2} \\ p_{21} & p_{22} & \cdots & p_{2r_2} \\ \vdots & \vdots & \ddots & \vdots \\ p_{r_1 1} & p_{r_1 2} & \cdots & p_{r_1 r_2} \end{pmatrix}$$

has rank one. The independence ideal $I_{1 \perp\!\!\!\perp 2}$ is generated by the 2×2 -minors:

$$I_{1 \perp\!\!\!\perp 2} = \langle p_{i_1 i_2} p_{j_1 j_2} - p_{i_1 j_2} p_{i_2 j_1} \mid i_1, j_1 \in [r_1], i_2, j_2 \in [r_2] \rangle.$$

For marginal independence, we already saw these quadratic binomial constraints in Chapter 1. $\square$

Figure 2: From the Lectures on Algebraic Statistics book

Algebraic varieties and polynomial ideals

We have already seen these in Lecture 1, and in homework 1 (problem 4). Here is a brief overview of what you need to know.

- A **variety** is the solution set to a simultaneous system of polynomial equations.
- If I is an **ideal**², then $V(I)$ is the variety defined by the vanishing of *all* polynomials in I .
- Hilbert basis theorem: even if I is infinite (it is!), there exists a **finite basis** for every I .

Important questions for statistics [digest next 2 slides]:

- what are points in a variety?
- how do you check if a point is on a variety?
- what if you are given an observation of 3 binary random variables, can you use polynomials to check some CI statements?

²an *ideal* is the infinite set of polynomial combination of some generating set.

Recall [Example 1.1.2](#) from the book: 3-step Markov chain.

$$p_{ijk} = P(X_1 = i, X_2 = j, X_3 = k) \text{ and } P(X_3 = k | X_1 = i, X_2 = j) = \frac{p_{ijk}}{p_{ij+}}.$$

You verified that a probability distribution, represented by a vector of probabilities $p = (p_{000}, p_{001}, p_{010}, p_{011}, p_{100}, p_{101}, p_{110}, p_{111}) \in \mathbb{R}^8$, being in this model is *equivalent* to the following four conditions:

$$p_{ijk} \geq 0 \text{ for all } i, j, k \in \{0, 1\}, \quad \sum_{i,j,k} p_{ijk} = 1,$$

$$p_{000}p_{101} - p_{001}p_{100} = 0, \text{ and } p_{010}p_{111} - p_{011}p_{110} = 0.$$

- In this example: **what is the variety?**

Recall [Example 1.1.2](#) from the book: 3-step Markov chain.

$$p_{ijk} = P(X_1 = i, X_2 = j, X_3 = k) \text{ and } P(X_3 = k | X_1 = i, X_2 = j) = \frac{p_{ijk}}{p_{ij+}}.$$

You verified that a probability distribution, represented by a vector of probabilities $p = (p_{000}, p_{001}, p_{010}, p_{011}, p_{100}, p_{101}, p_{110}, p_{111}) \in \mathbb{R}^8$, being in this model is *equivalent* to the following four conditions:

$$p_{ijk} \geq 0 \text{ for all } i, j, k \in \{0, 1\}, \quad \sum_{i,j,k} p_{ijk} = 1,$$

$$p_{000}p_{101} - p_{001}p_{100} = 0, \text{ and } p_{010}p_{111} - p_{011}p_{110} = 0.$$

- In this example: **what is the variety?**
- Is the point $(1/8, 1/8, \dots, 1/8)$ **on** this variety? (That is, **is this joint probability vector in the model?**)
 - [Verify right now, by hand.]

Recall [Example 1.1.2](#) from the book: 3-step Markov chain.

$$p_{ijk} = P(X_1 = i, X_2 = j, X_3 = k) \text{ and } P(X_3 = k | X_1 = i, X_2 = j) = \frac{p_{ijk}}{p_{ij+}}.$$

You verified that a probability distribution, represented by a vector of probabilities $p = (p_{000}, p_{001}, p_{010}, p_{011}, p_{100}, p_{101}, p_{110}, p_{111}) \in \mathbb{R}^8$, being in this model is *equivalent* to the following four conditions:

$$p_{ijk} \geq 0 \text{ for all } i, j, k \in \{0, 1\}, \quad \sum_{i,j,k} p_{ijk} = 1,$$

$$p_{000}p_{101} - p_{001}p_{100} = 0, \text{ and } p_{010}p_{111} - p_{011}p_{110} = 0.$$

- In this example: **what is the variety?**
- Is the point $(1/8, 1/8, \dots, 1/8)$ **on** this variety? (That is, **is this joint probability vector in the model?**)
 - [Verify right now, by hand.]
- Find an example of a point on the variety, which is a point in this model.
 - [Discussion of HW 1.b).]

Varieties

Points on the model $\leftrightarrow$ points on the variety.

What are model *ideals*?

If $p_{000}p_{101} - p_{001}p_{100} = 0$ and $p_{010}p_{111} - p_{011}p_{110} = 0$ both hold for a probability vector $p = (p_{000}, p_{001}, p_{010}, p_{011}, p_{100}, p_{101}, p_{110}, p_{111}) \in \mathbb{R}^8$, then **what other polynomial equations also hold?**

Varieties

Points on the model $\leftrightarrow$ points on the variety.

What are model *ideals*?

If $p_{000}p_{101} - p_{001}p_{100} = 0$ and $p_{010}p_{111} - p_{011}p_{110} = 0$ both hold for a probability vector $p = (p_{000}, p_{001}, p_{010}, p_{011}, p_{100}, p_{101}, p_{110}, p_{111}) \in \mathbb{R}^8$, then **what other polynomial equations also hold?**

- All such equations vanish on the points of the model.
 - [Discussion.]
- The (infinite) collection of all of these polynomials is **the ideal of this statistical model**.
- It is an equivalent description of the model.
 - [$\rightarrow$ algebraic geometry.]

Ideals

The (defining) ideal of a model is the set of all polynomials that simultaneously vanish on all points in the model.

Luckily, there is a finite basis for each ideal.

How to combine several CI statements?

Sum of ideals.

Example 3.1.10. Let X_1, X_2, X_3, X_4 be binary random variables, and consider the conditional independence model

$$\mathcal{C} = \{1 \perp\!\!\!\perp 3 \mid \{2, 4\}, 2 \perp\!\!\!\perp 4 \mid \{1, 3\}\}.$$

These are the conditional independence statements that hold for the graphical model associated to the four cycle graph with edges $\{12, 23, 34, 14\}$; see Section 3.2. The conditional independence ideal is generated by eight quadratic binomials:

$$\begin{aligned} I_{\mathcal{C}} &= I_{1 \perp\!\!\!\perp 3 \mid \{2, 4\}} + I_{2 \perp\!\!\!\perp 4 \mid \{1, 3\}} \\ &= \langle p_{1111}p_{2121} - p_{1121}p_{2111}, p_{1112}p_{2122} - p_{1122}p_{2112}, \\ &\quad p_{1211}p_{2221} - p_{1221}p_{2211}, p_{1212}p_{2222} - p_{1222}p_{2212}, \\ &\quad p_{1111}p_{1212} - p_{1112}p_{1211}, p_{1121}p_{1222} - p_{1122}p_{1221}, \\ &\quad p_{2111}p_{2212} - p_{2112}p_{2211}, p_{2121}p_{2222} - p_{2122}p_{2221} \rangle. \end{aligned}$$

Figure 3: From the Lectures on Algebraic Statistics book

Appendix

Here are some additional examples you may wish to explore, to familiarize yourself with conditional independence:

- [This](#) is where the Martin&Normal example came from.
- This informal [website](#) has some additional interesting examples.
- [Here is a set of slides](#) with several real-world examples of CI random variables.

License

This document is created for Math/Stat 561, Spring 2023 (2026 update), at Illinois Tech.

The first example is from Kaie Kubjas' course. Other online sources are cited throughout.

All materials posted on this page are licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](#).